

Gunratna Borkar

Dual Degree (B.Tech + M.Tech), IIT Bombay

✉ borkargunratna400@gmail.com |  Gunratna |  8605154284 |  Gunratna

Sr. AI Engineer at CAMS (India's largest RTA) - production-grade LLM/SLM fine-tuning, RAG pipelines, and document intelligence across investor triage, regulatory compliance, and GCP-native MLOps; serving 23 AMCs and Rs.47L Cr+ AUM.

PROFESSIONAL EXPERIENCE

CAMS | Sr. AI Engineer

[July '24 – Present]

► Email BOT | Multi-stage ML + LLM pipeline classifying and replying to investor emails

Processing **10k+** investor emails/day while automating **80%** of manual triage for 23 mutual funds

- **Junk Classifier:** Custom **PyTorch** residual network with sentence embeddings; **0.1% false negative rate** via 3-tier pipeline (Rules → ML Model → Semantic rescue), cutting false positives by **79%** (80%→16.8%) compared to rules
- **Intent Engine:** 4-bit **QLoRA** finetuned **Qwen-14B** (vLLM, **97%** macro-F1); migrated to **Gemini 2.5 Flash** with **JSON schema enforcement** extracting PAN, folios, ARNs and dates across **23 intents** × **5 sub-intents** eliminating GPU infra
- **Deployment:** Dual **FastAPI** microservices integrated with **Cloud Run** via **GitLab CI/CD**; **<500ms** p95 latency
Outcomes: 0.1% false negatives (Non Junk as Junk) · 79% FP reduction · 80% emails automated · **10k+** investor emails/day

► SEBI Debarred Entities | Regulatory Document Intelligence System

Evaluated on **6K+** SEBI orders, **50%** reduction in manual reviews via automated pipeline extracting debarred PAN holders

- **RAG Pipeline:** **FAISS**-based page relevance scoring on SEBI document patterns feeding context into **Gemini 2.5 Flash**
- **Document Routing:** **Multimodal** PDF routing using Camelot for text-based and Gemini Vision for scanned documents
- **Infrastructure:** Token-optimized and event-driven pipeline integrated with **Cloud Run**, **GKE**, **Pub/Sub** and **Firestore**
Outcomes: 50% reduction in manual reviews · schema-constrained outputs · 10 concurrent PDFs

► Other Projects

- **GitLab Agentic Pipeline** [Experimental]: 5-agent autonomous CI/CD system via **LangGraph** + **PydanticAI** — orchestrator decomposes PRDs → developer agent codes & raises **GitLab MRs** → reviewer + security agents coordinate via **MR comments** → auto-deploys post human review
- **Aadhaar PII Redaction:** Fine-tuned **CNN (ResNet/EfficientNet)** identifies Aadhaar pages in multi-page TIFFs; **YOLO** bounding-box detection + masking of first 8 of 12 Aadhaar digits for **DPDP Act** compliance
- **GST Invoice Extraction:** **RapidOCR** + **ONNX runtime** pipeline with parallel processing achieving **<1s** end-to-end latency; pattern-analysed regex engine extracting 7 structured fields from PDF/JPEG invoices

Vislesha Pvt Ltd | Data Science Intern

[July '23 – Sep '23]

Received a **Letter of Recommendation** from the **CTO** for exceptional analytical skills and steadfast dedication

- Built **30+** PySpark pipelines on **2M+** transaction records with **10×** speedup for dormancy prediction with **97%** accuracy
- Identified **5** key purchase segments via **FPGrowth** market basket analysis and **K-means/DBSCAN** clustering

TECHNICAL SKILLS

LLM & Agents: Fine-tuning (QLoRA), LangGraph, PydanticAI, LangChain, vLLM, RAG
ML / Vision: PyTorch, Tensorflow, ResNet, YOLO, ONNX Runtime, RapidOCR, Tesseract, Camelot
Cloud/MLOps: GCP (Cloud Run, GKE, Vertex AI, Pub/Sub, Firestore, GCS), Docker, GitLab CI/CD, FastAPI
Agentic AI Tools: Claude Code, GitHub Copilot Agent, OpenAI Codex, Gemini CLI, Google Antigravity

EDUCATION

IIT Bombay | Dual Degree (B.Tech + M.Tech), Metallurgical Engineering & Material Science

2019 – 2024

Overall CPI: **7.8/10** · M.Tech CPI (Year 5): **9.2/10**

Relevant Coursework: Statistical Machine Learning, Deep Learning, Linear Algebra, Probability & Statistics, Optimization

EXTRACURRICULAR ACTIVITIES

- **Internship, Edvizo** - Led sponsorship outreach, onboarded **10+** partners for NLCEE program targeting **10,000+** students
- **Teaching Assistant**, IIT Bombay - guided **120+** undergraduate students in a team of 6 through their labs and courses
- Mentored **4 students** at **Summer of Science** on a project focused on stock market analysis and investment strategies
- Completed a year-long course in **Indian Classical Music** under NSO Vocals, IIT Bombay